

XR-500 Industrial Controller

Operations & Maintenance Manual

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Target Model: XR-500 Series Control Systems

1. System Operational Limits & Parameters

1.1 Operating Envelope

The XR-500 system maintains automatic supervisory oversight of process variables. Operating outside defined envelope parameters triggers automatic warnings, throttling, or emergency trips.

Parameter	Normal Range	Maximum Limit	Units
Process Fluid Pressure	1.0 – 6.5	8.0	bar
Volumetric Flow Rate	10 – 120	150	L/min
Operating Temperature	-10 to 45	50	°C
Input Supply Voltage	20.0 – 28.0	32.0	VDC
System Duty Cycle	Continuous (24/7)	100% Load	—

2. Normal Operating Procedures

2.1 System Startup

1. Ensure the main electrical disconnect switch is placed in the **ON** position.
2. Verify system boot completion via the HMI display reading: **STATUS: READY**.
3. Press the blue **RUN** button on the control panel door or issue the **CMD_START** command viaModbus/TCP.
4. Verify system pressure stabilizes between 2.5 bar and 4.0 bar within 30 seconds of pump engagement.

2.2 Normal Shutdown Procedure

1. Disengage automatic sequence control on the supervisory workstation.
2. Issue the **CMD_STOP** command or hold the front panel **STOP** button for 3 seconds.
3. Allow the recirculation pump to run in flush mode for 120 seconds to cool internal drive elements.
4. Once fluid temperature drops below 35°C, close main inlet valves V-101 and V-102.

3. Preventive Maintenance Schedules

3.1 Periodic Maintenance Table

To maintain operational integrity and system compliance, perform scheduled maintenance tasks strictly according to the intervals listed below.

1.	+-----+-----+-----+		
2.	Interval	Task Description	Responsible
3.	+-----+-----+-----+		
4.	Daily	Visual inspection of status LEDs and display	Operator
5.	Monthly	Check pressure sensor calibration zero-points	Maintenance
6.	Every 6 Months	Clean/replace internal panel air filter	Maintenance
7.	Every 1,000 Hrs	Replace internal system battery (CR2032)	Lead Tech
8.	Every 12 Months	Full physical seal and valve diaphragm overhaul	Certified Tech
9.	+-----+-----+-----+		

3.2 Air Filter Replacement

1. Disconnect incoming 24 VDC power supply.
2. Remove the front enclosure filter cover by unscrewing the four corner thumb screws.
3. Remove the saturated filter element (Part No. FLT-XR500-A).
4. Insert the new filter element, ensuring airflow direction arrows point inward toward the heatsink assembly.
5. Re-secure cover thumb screws hand-tight.
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